

SHERIDAN, WY, USA

WMO: 726660

Lat: 44.769N

Lon: 106.969W

Elev: 1202

StdP: 87.69

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24029

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-20.0	-27.0	0.4	-23.2	-23.4	0.5	-19.8	14.6	-0.7	12.7	-0.2	1.6	310	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	35.1	17.4	33.3	17.1	31.5	16.8	19.8	30.1	18.8	29.2	17.9	28.4	3.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.4	23.9	15.2	12.5	22.8	14.1	11.6	22.1	61.5	30.1	58.1	29.0	55.0	28.3	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.9	9.0	DB	-28.5	38.3	3.3	1.5	-30.9	39.4	-32.8	40.3	-34.7	41.1	-37.0	42.2	(4)
(5)			WB	-28.6	21.9	3.0	1.5	-30.8	22.9	-32.5	23.8	-34.2	24.6	-36.4	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-4.2	-3.4	2.0	6.4	11.3	16.7	21.9	20.8	15.2	7.7	1.1	-4.2	(6)	
	DBStd	10.67	6.89	7.15	6.42	4.98	4.34	3.96	3.00	3.25	4.77	5.28	6.46	6.76	(7)	
	HDD10.0	2064	442	374	252	127	37	3	0	0	12	104	272	441	(8)	
	HDD18.3	4150	700	607	506	359	221	74	7	15	113	331	518	698	(9)	
	CDD10.0	1216	0	1	5	18	77	205	370	336	169	32	4	0	(10)	
	CDD18.3	260	0	0	0	0	2	26	118	93	21	1	0	0	(11)	
	CDH23.3	4476	0	0	2	18	94	522	1796	1481	514	48	1	0	(12)	
	CDH26.7	2063	0	0	0	1	18	196	909	723	204	12	0	0	(13)	
(14) Wind	WSAvg	3.1	2.9	3.2	3.5	4.0	3.7	3.1	2.7	2.6	2.7	3.0	2.8	2.9	(14)	
(15) Precipitation	PrecAvg	371	14	14	24	43	69	56	30	18	34	38	18	13	(15)	
	PrecMax	493	32	33	48	88	160	104	97	59	109	94	54	28	(16)	
	PrecMin	248	1	1	5	15	8	14	3	1	1	4	3	4	(17)	
	PrecStd	64	9	9	12	19	44	24	23	13	27	20	11	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.0	22.5	25.9	28.9	34.7	38.0	37.1	34.2	28.4	21.4	14.3	(19)	
		MCWB	5.7	7.1	9.7	12.2	15.6	17.2	18.0	17.9	16.2	13.4	9.5	5.6	(20)	
	2%	DB	10.0	12.4	18.5	22.5	25.9	31.4	35.6	34.9	31.7	24.0	17.3	10.2	(21)	
		MCWB	3.8	4.8	7.8	10.8	14.4	17.4	17.8	17.2	15.0	11.7	7.7	3.8	(22)	
	5%	DB	7.3	8.9	15.5	19.4	23.5	28.7	33.7	32.9	29.0	20.5	14.1	7.5	(23)	
		MCWB	2.4	3.1	6.9	9.7	13.8	17.0	17.5	17.0	14.5	10.5	6.3	2.3	(24)	
	10%	DB	4.9	6.1	12.5	16.4	21.0	26.3	31.8	31.0	26.3	17.1	10.7	4.8	(25)	
		MCWB	1.1	1.6	5.5	8.3	12.6	16.3	17.3	16.6	13.8	9.3	4.9	0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.6	10.2	13.2	17.5	20.6	21.0	21.5	18.4	14.1	10.1	6.1	(27)	
		MCDB	13.4	16.5	20.4	23.6	25.5	28.9	31.5	32.8	28.6	26.1	20.3	13.6	(28)	
	2%	WB	4.0	5.2	8.5	11.4	15.6	19.0	19.9	19.2	16.6	12.5	8.2	4.1	(29)	
		MCDB	9.6	11.9	17.8	20.6	23.7	27.7	30.6	30.5	27.7	22.4	16.4	9.7	(30)	
	5%	WB	2.6	3.4	7.2	10.1	14.3	17.9	19.0	18.1	15.5	11.0	6.6	2.7	(31)	
		MCDB	6.7	8.5	15.0	18.7	22.3	26.8	29.8	29.3	26.5	19.0	13.5	7.1	(32)	
	10%	WB	1.3	1.8	5.7	8.7	13.0	16.8	18.1	17.1	14.2	9.6	5.0	1.0	(33)	
		MCDB	4.6	5.6	11.9	15.8	20.4	25.4	29.0	28.3	24.7	16.9	10.5	4.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.1	13.3	13.6	13.9	15.3	17.8	17.9	16.9	14.6	13.5	12.1	(35)	
		MCDBR	14.5	15.7	19.0	19.8	19.1	20.2	21.3	21.7	22.0	20.8	18.7	15.0	(36)	
		MCWBR	9.3	9.6	10.2	10.4	9.9	9.3	7.8	8.3	9.0	10.6	10.6	9.6	(37)	
	5% WB	MCDBR	13.9	14.9	17.8	18.4	17.5	18.0	18.4	18.6	19.8	18.9	17.7	14.3	(38)	
		MCWBR	9.2	9.3	9.8	9.9	9.6	9.2	7.7	8.2	8.5	10.0	10.1	9.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.251	0.260	0.285	0.308	0.336	0.342	0.364	0.362	0.326	0.297	0.269	0.251	(40)	
	taud		2.534	2.515	2.493	2.446	2.434	2.445	2.415	2.396	2.488	2.536	2.529	2.527	(41)	
	Ebn at Noon		898	948	960	956	933	924	899	888	900	888	862	861	(42)	
	Edh at Noon		64	78	92	106	111	110	113	111	93	77	64	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.82	2.77	3.97	4.97	5.86	7.00	7.23	6.13	4.76	3.05	2.00	1.50	(44)	
	RadStd		0.11	0.14	0.27	0.28	0.59	0.51	0.28	0.27	0.34	0.25	0.16	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-0.55	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page